

GIT-HOL-3

Branching and Merging

Objective:

- Create a branch, make changes, merge with main/master.

Prerequisites:

- Completed HOL 2.

Procedure:

1. Create GitNewBranch.
2. Switch to the branch.
3. Create branch.txt.
4. Stage and commit changes.
5. Switch back to main.
6. Compare branches.
7. Merge GitNewBranch into main.
8. View log and delete merged branch.

Commands:

```
git branch GitNewBranch
git checkout GitNewBranch
echo "This file is created in GitNewBranch" > branch.txt
git add branch.txt
git commit -m "Added branch.txt"
git checkout main
git diff main GitNewBranch
git merge GitNewBranch
git log --oneline --graph --decorate
git branch -d GitNewBranch
git push origin main
```

OUTPUT:

```
MINGW64/c/Users/tavva/GitDemo
git checkout GitNewBranch
echo "This file is created in GitNewBranch" > branch.txt
git add branch.txt
git commit -m "Added branch.txt in GitNewBranch"
git status
git checkout main      # or master
git diff main GitNewBranch
git merge GitNewBranch
git log --oneline --graph --decorate
git branch -d GitNewBranch
git push origin mainbash: '$\E[200~cd': command not found
tavva@LAPTOP-304UFGD4 MINGW64 ~/GitDemo (main)
$
tavva@LAPTOP-304UFGD4 MINGW64 ~/GitDemo (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.
nothing to commit, working tree clean
tavva@LAPTOP-304UFGD4 MINGW64 ~/GitDemo (main)
$
tavva@LAPTOP-304UFGD4 MINGW64 ~/GitDemo (main)
$ git branch GitNewBranch
tavva@LAPTOP-304UFGD4 MINGW64 ~/GitDemo (main)
$
tavva@LAPTOP-304UFGD4 MINGW64 ~/GitDemo (main)
$ git branch
* main
  GitNewBranch
tavva@LAPTOP-304UFGD4 MINGW64 ~/GitDemo (main)
$ |
```

```
MINGW64/c/Users/tavva/GitDemo
tavva@LAPTOP-304UFGD4 MINGW64 ~/GitDemo (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.
nothing to commit, working tree clean
tavva@LAPTOP-304UFGD4 MINGW64 ~/GitDemo (main)
$
tavva@LAPTOP-304UFGD4 MINGW64 ~/GitDemo (main)
$ git branch GitNewBranch
tavva@LAPTOP-304UFGD4 MINGW64 ~/GitDemo (main)
$
tavva@LAPTOP-304UFGD4 MINGW64 ~/GitDemo (main)
$ git branch
* main
  GitNewBranch
tavva@LAPTOP-304UFGD4 MINGW64 ~/GitDemo (main)
$ git checkout GitNewBranch
Switched to branch 'GitNewBranch'
tavva@LAPTOP-304UFGD4 MINGW64 ~/GitDemo (GitNewBranch)
$ git branch
* GitNewBranch
  main
tavva@LAPTOP-304UFGD4 MINGW64 ~/GitDemo (GitNewBranch)
$ echo "This file is created in GitNewBranch" > branch.txt
tavva@LAPTOP-304UFGD4 MINGW64 ~/GitDemo (GitNewBranch)
$ cat branch.txt
This file is created in GitNewBranch
tavva@LAPTOP-304UFGD4 MINGW64 ~/GitDemo (GitNewBranch)
$ git add branch.txt
warning: in the working copy of 'branch.txt', LF will be replaced by CRLF the next time Git touches it
tavva@LAPTOP-304UFGD4 MINGW64 ~/GitDemo (GitNewBranch)
$ git commit -m "Added branch.txt in GitNewBranch"
[GitNewBranch 37b5822] Added branch.txt in GitNewBranch
1 file changed, 1 insertion(+)
create mode 100644 branch.txt
tavva@LAPTOP-304UFGD4 MINGW64 ~/GitDemo (GitNewBranch)
```

```

tavva@LAPTOP-J04UFCD4 MINGW64 ~/GitDemo (GitNewBranch)
$ git status
On branch GitNewBranch
nothing to commit, working tree clean

tavva@LAPTOP-J04UFCD4 MINGW64 ~/GitDemo (GitNewBranch)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

tavva@LAPTOP-J04UFCD4 MINGW64 ~/GitDemo (main)
$ git merge GitNewBranch
Updating 4e82564..37b6882
Fast-forward
 branch.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 branch.txt

tavva@LAPTOP-J04UFCD4 MINGW64 ~/GitDemo (main)
$ git log --oneline --graph --decorate
* 37b6882 (HEAD -> main, GitNewBranch) Added branch.txt in GitNewBranch
* 4e82564 (origin/main) Added gitignore file
* f6754c9 Added welcome file

tavva@LAPTOP-J04UFCD4 MINGW64 ~/GitDemo (main)
$ git branch -d GitNewBranch
Deleted branch GitNewBranch (was 37b6882).

tavva@LAPTOP-J04UFCD4 MINGW64 ~/GitDemo (main)
$ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 366 bytes | 366.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/tavvabhanu/GitDemo
 4e82564..37b6882  main -> main

```

Result:

Branch was successfully merged into the main branch.